MILITARY SPECIFICATION SHEET

FILTER ELEMENT, FLUID PRESSURE, HYDRAULIC, AIRCRAFT, DISPOSABLE, 5 MICRON ABSOLUTE

This specification sheet is approved for use by Code 99, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the filter elements described herein shall consist of this document and the latest issue in effect of Specification MIL-F-83860.

REQUIREMENTS:

Bubble Point. The initial and burst out bubble point values shall meet the values established by the activity responsible for qualification and shall be determined as specified in 4.6.1 of general specification.

Filtration. The filter elements shall remove 100 percent of all contaminants over 5 microns in size and 95 percent by weight of add contaminants.

Cold Start. The filter elements shall withstand 10 flow cycles as specified in 4.6.6 of general specification at -65°F.

Element Pressure Drop. The pressure drop through the clean element using MIL-H-5606 hydraulic fluid at 100 ± 10°F shall not exceed 10 psid at rated flow.

Element Dirt Capacity. The filter elements shall meet the dirt capacities specified in Table I, Contaminants.

Element Collapse. The filter elements shall withstand the following differential pressures:

Part Number	Differential Pressure
M83860/1-6	4500 psid
M83860/1-8	4500 psid
M83860/1-10	4500 psid
M83860/1-12	4500 psid
M83860/1-16	4500 psid

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Description. The filter elements shall be as shown in Figure 1 and specified herein.

Flow Fatigue. The filter elements shall withstand the flow fatigue cycles specified at 300 cpm:

Cycles	Pressure Drop
1,120,000	20
112,000	50
12,500	100
400	1500
100	3000

Media Migration. There shall be no media migration traceable to the filter elements when tested in MS28896 housing, at vibration frequencies specified in 4.6.9 of general specification.

Maximum Particle Passed. The maximum particle passed shall be no greater than 5 microns.

TABLE I. CONTAMINANTS

		Degree of Filtration Test		Maximum Particle Test	Pressure Buildup Test	
Dash No.	Flow Rate (gpm)	Add Rate F-9 Beads (gms)	Add Rate AC Dust (gms)	Add Rate Carbonyl Iron E (gms)	Add Rate AC Dust (gms)	Minimum AC Dust to 90 psid (gms)
-6	3.5	0.15	0.15	0.30	0.10	1.4
-8	6.0	0.15	0.15	0.30	0.10	1.8
-10	10.5	0.15	0.15	0.30	0.18	3.4
-12	16.0	0.15	0.15	0.30	0.26	5.2
-16	29.0	0.15	0.15	0.30	0.49	7.4

Part Number. The part number consists of an "M" plus the basic number of this specification sheet and a dash number.

Example: M83860/1-6

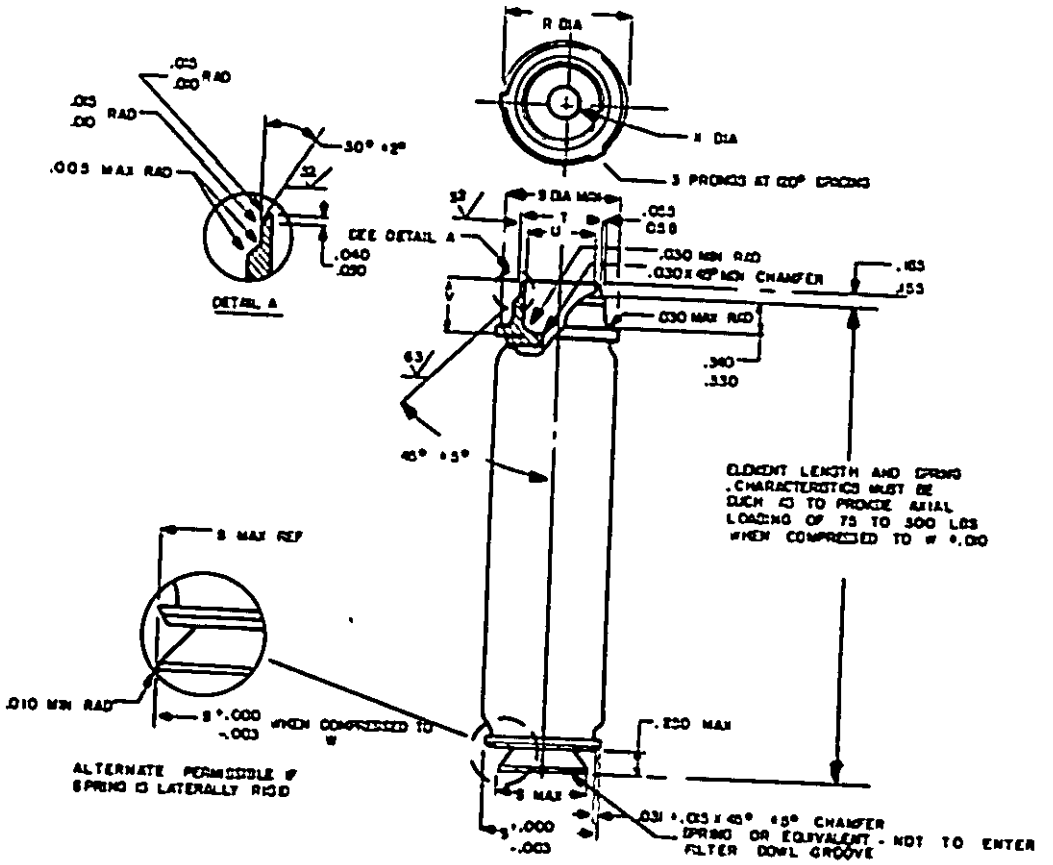
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DASH NO.	RATED MAX FLOW GPM	R +.000 -.010	S	T ±.001	U MIN	V MIN	W	X MIN
-6	3.5	1.250	1.137	.748	.670	.490	3.625	.435
-8	6.0	1.250	1.137	.748	.670	.490	4.095	.435
-10	10.5	1.415	1.290	.875	.790	.600	5.235	.555
-12	16.0	1.625	1.500	1.122	1.035	.690	6.235	.715
-16	29.0	1.985	1.845	1.372	1.155	.760	7.235	.995

FINISH: 100 ALL OVER EXCEPT AS NOTED, IN ACCORDANCE WITH ANSI B46.1 - 1971.
DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.010, ANGLES ±1°.
CONCENTRICITY: SURFACE OF DIAMETERS T, R AND S MUST BE CONCENTRIC WITHIN .012 TOTAL INDICATOR READING.

FIGURE 1

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